

Ammar Alam

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Work Experience

- **Software Engineer** – Stellic Jul '22 – Present
 - Led complete redesign of course scheduling engine alongside senior algorithm engineer, architecting orchestration layer and combination-generation logic around cutting-edge local search algorithms
 - Delivered 10x performance gains while dramatically improving schedule quality and student satisfaction, replacing legacy genetic algorithm system
 - Engineered mission-critical features—General Requirements, Test Scores, Scheduler powering degree management for 70+ elite universities including MIT and CMU, impacting 1M+ students globally
 - Pioneered General Requirements as first fully Clean Architecture implementation with 70
 - Transformed backend performance across Python/Django REST services through strategic database optimization, bulk operations, intelligent prefetching, and async processing via Celery—achieving 10-1000x improvements in API response times
 - Championed enterprise-wide migration to Clean Architecture following comprehensive research into modern software patterns, elevating test coverage from <3
 - Accelerated feature delivery while significantly enhancing code maintainability through Clean Architecture implementation
 - Established team-wide excellence standards through Test-Driven Development and Mob Programming training programs, streamlining code review processes and elevating engineering productivity across the organization
- **Data Engineering Intern** – Systems Ltd. Feb '22 – Apr '22
 - Pioneered machine learning-driven Robotic Process Automation solutions, revolutionizing client business workflows through intelligent automation
 - Engineered high-performance ETL pipelines leveraging Python, Pandas, and Selenium—delivering 90
- **Computer Vision Intern** – The Sparks Foundation Jun '21
 - Conducted comprehensive performance analysis of face detection algorithms, implementing both traditional Haar Cascades and state-of-the-art deep learning approaches using Keras
 - Developed innovative dataset fusion methodology combining multiple face mask datasets, achieving industry-leading 99

Technical Skills

- **Languages:** Python, Bash, JavaScript, C++, SQL, Verilog, MATLAB
- **Frameworks/Libraries:** Django, React, OpenCV, TensorFlow, WebGL, CUDA, NLTK, Pandas, Matplotlib
- **Infrastructure/Cloud:** AWS (EC2, S3, EBS, ECR), Docker, Redis, GitHub Actions
- **Monitoring/Logging:** Grafana/Loki, Datadog, Sentry
- **Collaboration/Tools:** Git, Postman, OpenAPI, Notion, Asana

Education

- **Habib University**, Pakistan Aug '18 – Jun '22
 - B.Sc. in Computer Science, Minor in Electrical & Computer Engineering, Minor in Mathematics
 - CGPA:** 3.76/4.00

Honors, Awards, and Memberships

- Member, ACM; Vice Chair, ACM Habib Chapter
- 60% Academic Scholarship Habib University
- President's List: Consecutive appearances in Dean's List
- Dean's List: Top 10% of department Spring '19, Fall '19, Fall '20

Publications

- [1] **S. M. A. Alam**, M. U. Huda and M. Farhan, "[Alpha-trimmed Mean Filter and XOR based Image Enhancement for Embedding Data in Image](#)", 2021 International Conference on Visual Communications and Image Processing (VCIP), Munich, Germany, 2021, pp. 1-4, doi: 10.1109/VCIP53242.2021.9675315.